D.A.V. PUBLIC SCHOOL, SECTOR IV, B. S. CITY

CLASS XI — MATHEMATICS REVISION SHEET

UNIT TEST III COMBINED QUESTION BANK **TOPIC-WISE STRUCTURED COMPILATION**

GENERAL INSTRUCTIONS:

1. This question bank categorizes problems topic-wise from Unit Test III (Set A & Set B) along with essential variations for thorough concept review.
2. Work through each topic sequentially to gain complete mastery over Straight Lines, Conics, and 3D Geometry concepts.

UNIT 1: STRAIGHT LINES

Q1. The general equation of a straight line in two variables is represented as:

(a) $2x + 3y = a$ (b) $ax + by + 0 = 1$

(c) $ax + by + c = 0$ (d) $x/a + y/b = 1$

Q2. Calculate the distance between the given parallel lines:

(i) $3x - 4y + 7 = 0$ and $3x - 4y + 5 = 0$

(ii) $3x - 4y + 12 = 0$ and $3x - 4y + 2 = 0$

(iii) $15x + 8y + 31 = 0$ and $15x + 8y - 34 = 0$

(iv) $l(x + y) + p = 0$ and $l(x + y) - r = 0$

Q3. By using the concept of the equation of a line, prove that the following sets of points are collinear:

(i) (1, 1), (3, 5) and (5, 9)

(ii) (3, 0), (−2, −2) and (8, 2)

(iii) (a, 0), (0, b) and (1, 1) if $(1/a + 1/b = 1)$

Q4. Image of a Point with respect to a Line:

(i) Find the image of the point (1, 2) with respect to the line $2x + y = 4$ assuming the line to be a plane mirror.

(ii) Find the image of the point (3, 8) with respect to the line $x + 3y = 7$ assuming the line to be a plane mirror.

(iii) Find the coordinates of the foot of the perpendicular from (−1, 3) to the line $3x - 4y - 16 = 0$.

Q5. Perpendicular Distance & Intercept Theorems:

(i) If p is the length of the perpendicular from the origin to the line whose intercepts on the axes are a and b, then show that $1/p^2 = 1/a^2 + 1/b^2$.

(ii) Prove that the product of the lengths of the perpendiculars drawn from the points $(\pm\sqrt{(a^2 - b^2)}, 0)$ to the line $(x/a)\cos\theta + (y/b)\sin\theta = 1$ is equal to b^2.

(iii) Find the equation of the line passing through (−3, 5) and perpendicular to the line through the points (2, 5) and (−3, 6).

UNIT 2: CONIC SECTIONS — CIRCLE & PARABOLA

Q6. Find the centre and radius of each of the following circles:

(i) $x^2 + y^2 - 6x + 4y - 12 = 0$

(ii) $x^2 + y^2 - 8x + 10y - 12 = 0$

(iii) $2x^2 + 2y^2 - x = 0$

Q7. Standard Parabolas & Latus Rectum:

(i) The length of the latus rectum of the parabola $y^2 = 16x$ is:

(a) 4 (b) 8

(c) 12 (d) 16

(ii) The length of the latus rectum of the parabola $y^2 = 12x$ is:

(a) 13 (b) 12

(c) 11 (d) 10

(iii) Find the focus, axis, equation of directrix, and latus rectum of $x^2 = -9y$.

UNIT 2 (CONTD.): CONIC SECTIONS — ELLIPSE & HYPERBOLA

Q8. Standard Equations of Conics:

(i) The standard equation of a hyperbola with transverse axis along x-axis is given by:

(a) $x^2/a^2 + y^2/b^2 = 1$ (b) $x^2/a^2 - y^2/b^2 = 1$

(c) $x^2 = y^2$ (d) $x^2 + y^2 = a^2$

Q9. Find the standard equation of the ellipse satisfying the following conditions:

(i) Vertices are $(\pm 10, 0)$ and foci are $(\pm 6, 0)$

(ii) Vertices are $(\pm 13, 0)$ and foci are $(\pm 5, 0)$

(iii) Ends of major axis are $(0, \pm\sqrt{5})$ and ends of minor axis are $(\pm 1, 0)$

(iv) Length of major axis 20 and foci at $(0, \pm 5)$

Q10. Locus & Applied Real-Life Problems on Conics:

(i) A rod AB of length 15 cm rests between two coordinate axes such that end A lies on the x-axis and end B lies on the y-axis. A point P(x, y) is taken on the rod such that AP = 6 cm. Show that the locus of P is an ellipse.

(ii) A beam is supported at its ends by supports which are 12 m apart. Since the load is concentrated at its centre, there is a deflection of 3 cm at the centre and the deflected beam is in the shape of a parabola. How far from the centre is the deflection 1 cm?

(iii) An arch is in the form of a parabola with its axis vertical. The arch is 10 m high and 5 m wide at the base. How wide is it 2 m from the vertex of the parabola?

UNIT 3: INTRODUCTION TO THREE DIMENSIONAL GEOMETRY

Q11. Fundamental Coordinate Geometry in Space:

(i) The y-coordinate of any point lying on the ZX-plane is:

(a) 1 (b) −1

(c) 0 (d) Any non-zero number

(ii) The coordinate planes divide the entire 3D space into _____ octants:

(a) 3 (b) 2

(c) 8 (d) 4

(iii) Find the locus of a point for which $z = 0$ and $x = 0$ simultaneously.

(iv) State the octant in which the points $(-3, 1, 2)$, $(-2, -4, -7)$, and $(4, -2, 3)$ lie.

Q12. Distance Formula & Section Formula in 3D Space:

(i) Find the distance between the points $P(1, -3, 4)$ and $Q(-4, 1, 2)$.

(ii) Show that the points $A(1, 2, 3)$, $B(-1, -2, -1)$, $C(2, 3, 2)$ and $D(4, 7, 6)$ are the vertices of a parallelogram.

(iii) Find the coordinates of the point which divides the line segment joining the points $(1, -2, 3)$ and $(3, 4, -5)$ in the ratio 2 : 3 internally.

(iv) Using section formula, prove that the points $P(2, -3, 4)$, $Q(-1, 2, 1)$ and $R(0, 1/3, 2)$ are collinear.